

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 23: Search and Rescue (Annex 12)

Syllabus Source: CAE Oxford (pp. 471-482) | Pass Target: >= 90% AviationExam

1. SAR ORGANIZATION & RESCUE COORDINATION CENTRES (RCC)

ICAO Annex 12 mandates that Contracting States establish Search and Rescue Services on a 24-hour basis. A Rescue Coordination Centre (RCC) is established in each Search and Rescue Region (SRR) to direct SAR operations.

2. GROUND-AIR VISUAL SIGNAL CODE FOR SURVIVORS

Ground signals made by survivors to search aircraft using fabric strips, wood, or trodden snow:

Visual Symbol	Meaning to Search Aircraft (Annex 12 Appendix 1)
V	Require assistance
X	Require medical assistance
N	NO or Negative
Y	YES or Affirmative
-> (Arrow)	Proceeding in this direction
LL	All well / All survivors located
JL	Operation completed

[!] EXAM TRAP (AVIATIONEXAM) - Search Aircraft Acknowledgment Signals

When a search aircraft observes survivor ground signals: - DAY: Rocking wings (for aeroplanes) or flashing landing lights twice (for helicopters). - NIGHT: Flashing landing lights or navigation lights TWICE. - If signal NOT understood: Aeroplane completes a 360° turn to the right, or displays red pyrotechnics.

3. DROPPABLE EMERGENCY EQUIPMENT COLOR STREAMERS

Containers or packages containing survival equipment dropped to survivors are indicated by colored streamers: - RED: Medical supplies and first aid equipment. - BLUE: Food and water. - YELLOW: Blankets and protective clothing. - BLACK: Miscellaneous survival equipment (knives, compasses, cooking equipment).

4. EMERGENCY LOCATOR TRANSMITTER (ELT) FREQUENCIES

- 406.0 MHz: Digital satellite frequency monitored by COSPAS-SARSAT satellite constellation. Transmits unique 24-bit aircraft hex code for instant identification. - 121.5 MHz: VHF civilian emergency frequency used for terminal acoustic/radio homing by rescue aircraft.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does the ground-air survivor symbol 'X' signify to search aircraft?

- [A]: Require assistance
- [B]: Require medical assistance
- [C]: No or Negative
- [D]: Aircraft destroyed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The single 'V' means 'Require assistance'; the 'X' means 'Require medical assistance'.

Q2: How does a search aeroplane acknowledge that survivor ground signals have been seen and understood?

- [A]: Firing a red flare
- [B]: Rocking wings
- [C]: Circling to the right 360°
- [D]: Pitching nose up and down

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Rocking wings by day (or flashing lights twice at night) confirms signals are understood.

Q3: Which satellite frequency is used by modern ELTs for global satellite alerting through COSPAS-SARSAT?

- [A]: 121.5 MHz
- [B]: 243.0 MHz
- [C]: 406.0 MHz
- [D]: 1090 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. 406.0 MHz is the digital satellite beacon frequency (121.5 MHz is for VHF homing).

Q4: What color streamer indicates that a dropped survival package contains food and water?

- [A]: Red
- [B]: Blue
- [C]: Yellow
- [D]: Black

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under Annex 12, Blue denotes food and water (Red = medical, Yellow = blankets/clothing, Black = miscellaneous).

Q5: What does the ground-air visual signal 'LL' signify?

- [A]: Require food and water
- [B]: All well
- [C]: Land here
- [D]: Lost direction

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 'LL' indicates 'All well'.

Q6: What is the international military UHF aeronautical emergency frequency monitored by SAR units?

- [A]: 121.5 MHz
- [B]: 243.0 MHz
- [C]: 406.0 MHz
- [D]: 1090 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 243.0 MHz is the international UHF military aeronautical emergency frequency.

Q7: What does the ground-air survivor visual symbol 'V' signify to search aircraft?

- [A]: Require medical assistance
- [B]: Require assistance
- [C]: All well
- [D]: Operation completed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Single 'V' signifies 'Require assistance' ('X' is require medical assistance; 'LL' is all well).